INCH-POUND

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COMMERCIAL ITEM DESCRIPTION

ANALYZER, SPECTRUM, 2.9 GHZ RF

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID) describes the general requirements for a high-performance RF spectrum analyzer. This CID is meant as a minimum requirement for the 2.9 GHz RF Spectrum Analyzer in which only those manufacturers that meet or surpass the following requirements are to be supplied with per this. The RF spectrum analyzer is also referenced as the equipment.

2. CLASSIFICATION. The RF spectrum analyzer covered by this CID shall be commercially available equipment that has factory-installed options and/or may be modified to the extent necessary to meet the following description. The RF spectrum analyzer shall be class 3, in accordance with MIL-PRF-28800.

3. SALIENT CHARACTERISTICS.

3.1 Item description. RF spectrum analyzer.

3.2 Frequency range. The equipment shall have a minimum operating frequency range from 9 kHz to 2.9 GHz.

3.3 Reference aging rate. The equipment frequency reference aging rage shall be $\leq \pm 2 \bullet 10^{-6}$ per year and power-on stability deviation after 24 hours on time $\leq 5 \times 10^{-6}$.

3.4 Frequency span range. The equipment frequency span range of the equipment shall be 0 Hz (zero span), 100 Hz to 3 GHz.

3.5 Frequency span accuracy. The equipment frequency accuracy shall be within ±1.0% of span.

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3.6 Frequency span resolution. The equipment shall have a resolution span of 2 Hz.

3.7 Sweep time range. The equipment shall have a range of 4ms to 4000s.

3.8 Sweep time accuracy. The equipment shall have a seep time accuracy of ±1%.

3.9 Sweep trigger. The equipment shall be capable of being triggered from free run, single, line, video, external, delayed, offset.

3.10 Resolution bandwidth range. The resolution bandwidth range settings for the equipment shall have a range of 1 kHz to 3 MHz and -3 dB bandwidth.

3.11 Resolution bandwidth accuracy. The resolution bandwidth accuracy for the equipment shall be within ±15%, for the range of 1 kHz to 3 MHz: ±20%, for the range of 9 kHz to 120 kHz.

3.12 Resolution bandwidth selectivity. The resolution bandwidth selectivity shall be, for a 60 dB/3 dB bandwidth ratio; 1 kHz to 5 MHz RBW.

3.13 Video bandwidth range. The equipment shall have a video bandwidth range of 30 Hz to 1 MHz in 1-3-10 sequence.

3.14 Measurement range. The equipment shall have a Displayed Average Noise Level (DANL) to +30 dBm.

3.15 Input attenuator range. The equipment shall have a 0 to 65 dB, in 5 dB steps.

3.16 Displayed Average Noise Level (DANL). The equipment shall be ≤ -117 dBm (10 MHz to 1.0 GHz, 1kHz RBW).

3.17 Frequency amplitude. The equipment shall have an overall amplitude accuracy of ±1.1 dB.

3.18 Third Order Intermodulation distortion (TOI). The equipment shall be +7.5 dBm (100 MHz to 3 GHz).

3.19 Design and construction. The equipment shall be used in both bench-top and rack-mount environments. The equipment is a commercial item subject to commercial-product regulatory standards. The equipment shall be designed and constructed in accordance with all applicable European Union (EU) requirements with the “CE” marking affixed, International Organization for Standardization (ISO), the International Electro-technical Commission (IEC), the International Telecommunication Union (ITU), Underwriters Laboratories (UL), and the American National Standards Institute (ANSI). The equipment shall be compliant with the Restriction of Use of Hazardous Substances (RoHS) and Waste from Electrical and Electronic Equipment (WEEE) guidelines and policies. It shall be designed and constructed so that no parts will work loose in service, and to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.20 System safety. The equipment shall comply with all sections of the UL 1449 standard for electrical safety and the IEC 61010-1.

3.21 Sound levels. The equipment acoustic noise shall conform to sound-level requirements of ISO 7779 and measured in accordance with ISO 8253, section 2.

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3.22 Electrostatic Discharge (ESD). The design of the equipment shall preclude equipment damage due to ESD, protect personnel from electrical shock due to static charging, and prevent ignition of explosive atmospheres due to sparking in accordance with IEC 61340-5-1:2024 section 5.2.

3.23 Electromagnetic Interference (EMI). The equipment's EMI emissions shall be in accordance with the radiated-emission and susceptibility requirements of MIL-PRF-28800; MIL-STD-461, sections RE102 and RS103; IEC 1000-2-5; and the emitted radiation absorption and radiation limits of CFR, Title 47, Parts 15.109 and 15.209.

3.24 Service life. The design of the equipment shall be such that, under normal use and operation, the service repair can be performed according to MIL-PRF-28800, sections 3.15.2 and 3.15.3.

3.25 Port damage. The input ports shall withstand up to 12 volts root-mean-square (V_{RMS}) for inputs A and B without damaging the internal circuitry. The input port "C" shall withstand up to 2.5 V_{RMS} without damaging the internal circuitry.

3.26 Connectors. The equipment shall have be a precision type "N" female connector, external trigger input, BNC female connector, audio output connector compatible with standard earphones, video output connector. A general-purpose-interface-bus (GPIB) interface with an IEC 60488 (formerly IEEE-488.2) protocol connector is required for compatibility and control functions.

3.27 Operating-temperature range. The equipment shall meet the specifications listed herein with full capabilities within the temperature range of 32 to 131 °F (0 to 55 °C).

3.28 Storage-temperature range. The equipment shall meet or exceed manufacturer's specifications for stored equipment within the temperature range of -40 to 168.8 °F (-40 to 75 °C) after tests of MIL-PRF-28800, section 4.5.5.1, have been performed for instruments in the range of class 3 equipment classification per MIL-PRF-28800, Table XI.

3.29 Operating-humidity range. The equipment shall meet its performance and accuracy requirements for humidity conditions as specified in MIL-PRF-28800, section 3.9.2.3 within the humidity range of 5%-95% 50 to 86 °F (10 to 30 °C).

3.30 Altitude requirements. The equipment shall meet its operating performance and accuracy while operating at an altitude of 15,000 feet (4,600 meters). The equipment shall meet its performance and accuracy requirements for operating and non-operating storage as specified in MIL-PRF-28800, sections 3.9.3 and 4.5.5.2 respectively, at this altitude.

3.31 Heat dissipation and ventilation. Ventilation shall be sufficient to allow for proper heat dissipation that can be achieved by either passive or active methods in order to meet MIL-PRF-28800, section 4.5.5.1, under temperature and humidity tests for class 3 equipment classification per MIL-PRF-28800, Table XI.

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3.32 Calibration and maintenance adjustments. No special tools, software, technical manuals, or documentation shall be required for the alignment, adjustment, repair, and calibration of the equipment, except those provided with the equipment. A software license shall be issued for each physical instrument when the system is designed for computer-based calibration. Software licenses needed to perform the alignment/adjustment process (where required by the manufacturer) shall be valid for the life of the instrument and shall be included with the equipment. Any special or unique accessories not common to a secondary-calibration laboratory required for calibration of this item shall be furnished by the manufacturer, to include, but not limited to, software, fixtures, tools, and standards.

3.33 Calibration period. The design of the equipment shall provide readily accessible calibration and maintenance adjustments. The calibration adjustments shall be accessible without removal of the test-set case or modules. The calibration interval shall be based on a period of one calendar year or an operating interval of 2,080 hours, whichever occurs first. At the end of this interval, 85 percent of the units shall remain within calibration tolerances in accordance with MIL-PRF-28800, section 3.13.

3.34 Reliability. The design of the equipment shall be such that under normal use and operation, the mean time between failures (MTBF) shall exceed 2,080 hours of operation with a statistical certainty equal to or greater than 95 percent in accordance with MIL-PRF-28800, section 3.14, based on an 8-hour workday, 40-hour week, and 52-week interval. The manufacturer shall be able to reasonably establish the claim and certify with NIST traceability that the offered equipment is capable of the required level of reliability by meeting or exceeding MIL-PRF-28800, sections 3.14 and 4.5.8.

3.35 Built-in test equipment and diagnostics. The equipment shall be equipped with built-in test and diagnostic capability that performs a system test evaluating all functional characteristics of the instrument at “Power On” or “Cold Start” in accordance with MIL-PRF-28800, section 6.6.6.

3.36 Maintainability. The equipment shall meet the human-engineering factors of ergonomics for operational and maintainability criteria providing access to controls, internal components, and adjustments in accordance with MIL-STD-1472, sections 5.9 through 5.9.16, and MIL-PRF-28800, section 6.6.15.

3.37 Preventive maintenance. Preventive maintenance shall not require more than 15 minutes per 30-day period. Preventive maintenance shall not require breaking of the equipment seams where calibration seals would normally be placed in accordance with MIL-PRF-28800, section 3.15.2.

3.38 Corrective maintenance. The equipment shall be designed to meet reparability criteria in accordance with MIL-PRF-28800, section 3.15.1.

3.39 Electrical power sources and connections. The equipment shall operate from commercial, military, and shipboard power sources with a nominal voltage of 120 voltage root-mean-square (V_{RMS}), single-phase alternating current (AC) (±10%) at line frequencies of 50 Hz and 60 Hz, in accordance with MIL-PRF-28800, sections 3.6.1.2 and 3.6.1.2a.

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3.40 Electrical surge. The equipment shall be able to provide surge protection that meets UL 1449, section 38; IEC 61000-4-4; and IEC 61010-1.

3.41 Noise suppression. The equipment shall be able to operate with electrical noises and harmonic content present on the alternating-current power source with frequencies in the 30 Hz to 50 kHz range and levels varying from line level of 41.6 dB ±1 dB (108 to 132 VAC) measured at the power input in accordance with UL 1283, IEC/EN 60939-1 and under the guidance of the IEEE-519 standard.

3.42 Dimensions. The equipment shall comply with the width and depth of the EIA/ECA-310-E standard equipment rack configuration.

3.43 Rack mount. The equipment shall be capable of being rack mounted in a standard EIA/ECA-310-E equipment rack and, if required, include a rack-mount conversion kit that meets the requirements of MIL-PRF-28800, section 3.7.4.13.3.

3.44 Weight. The total weight of the equipment, excluding accessories and manuals, shall comply with MIL-STD-1472 for weight and size restrictions for single-person lift and handling, and shall not exceed a one-person lift ≤ 35 pounds (15.6 kilograms).

3.45 Markings. All warning decals and instrument-category ratings must conform to commercial safety standards and be visible by the user in accordance with commercial product-safety labeling regulations and MIL-STD-130. Markings shall be applied with paint, decals, engraving, or molding, and shall be the largest size possible. Warning/Danger text shall be lusterless red.

4. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials or bio-based content to the maximum extent practicable, in accordance with Federal Acquisition Regulation, Part 23.403.

5. PRODUCT CONFORMANCE. Verification of the delivered equipment set will be performed by the end user at the delivery point. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Sources of documents.

7.1.1 Government Documents. Copies of Military Specifications, Standards, and Handbooks are available online at https://quicksearch.dla.mil.

7.1.2 Code of Federal Regulations (CFR). Copies of the CFR are available online at https://www.ecfr.gov.

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7.1.3 Electronic Components Industry Association (ECIA). Copies of the ECIA are available online at https://www.ecianow.org.

7.1.4 EOS/ESD Association. Copies of the EOS/ESD are available online at https://www.esda.org.

7.1.5 International Electrotechnical Commission (IEC). Copies of the ISO/IEC are available online at https://www.iec.ch.

7.1.6 Institute of Electrical and Electronics Engineers (IEEE). Copies of the IEEE are available online at https://www.ieee.org.

7.1.7 Underwriters Laboratories (UL). Copies of the UL are available online at https://www.ul.com.

7.1.8 Federal Acquisition Regulation (FAR). Copies of the FAR are available online at https://www.acquisition.gov.

7.2 Ordering data. Procurement documents must specify title, number, date of this CID, the issue of the ASSIST Online (https://assist.dla.mil) to be used in the solicitation, preservation and packaging requirements.

7.3 Key words.

Altitude
Built-in test
Calibration
Frequency
Maintainability

MILITARY INTEREST

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